

CHERISH VASANT

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PROFILE

I'm a final-year CSE student, interested in AI, ML, Deep Learning, NLP, Generative AI, and Agentic AI systems. I enjoy understanding how these models work and how they can be used to solve real-world problems. I love problem solving, and I learn best by building projects and experimenting with ideas. I'm always trying to go deeper, improve my skills, and turn what I learn into something practical.

EDUCATION

Vellore Institute of Technology

B.Tech in Computer Science and Engineering (spec. AI & ML)

- CGPA: 9.24 / 10.0

Chennai, India

Aug 2023 – May 2027

Sishya OMR

Grade 12 – ISC, Science with Computer Science – 96%

Chennai, India

2021 – 2023

St. Patrick's High School

Grade 10 – CISCE – 95.8%

Chennai, India

2019 – 2021

PROJECTS

TrebleAI | AI-Powered Music Learning Platform, AI Agents, Full-Stack

- Designed and developed **TrebleAI**, an AI-powered self-learning music platform that enables users to independently learn and practice sheet music through intelligent automation, reducing reliance on instructors and tutorial videos.
- Engineered an end-to-end music processing pipeline using Audiveris, music21, MIDI generation, and FluidSynth to convert uploaded sheet music into synchronized audio and interactive notation.
- Built AI-powered tutoring agents using LangChain with multi-tool integration and structured JSON outputs to answer questions about uploaded scores, retrieve music theory, and provide personalized learning assistance.
- Architected a full-stack application using FastAPI background tasks, Next.js, TypeScript, and PostgreSQL, integrating AI tutoring, intelligent search, synchronized sheet music rendering, animated piano visualization, and real-time playback into a unified platform.

Cloud Security for Remote Patient Monitoring Systems | Machine Learning, Cybersecurity, IoT

- Developed a multi-class intrusion detection system using the **Edge-IIoTset** dataset to classify normal traffic and **14 cyberattack categories**, including DDoS, DoS, Backdoor, SQL Injection, Cross-Site Scripting (XSS), Command Injection, Password attacks, MITM, Port Scanning, Ransomware, Fingerprinting, Uploading attacks, Vulnerability Scanning, and Botnet activity.
- Built a preprocessing and feature engineering pipeline, reducing the feature space to the **30 most informative features**, and benchmarked XGBoost, CatBoost, Random Forest, Gradient Boosting, Decision Tree, KNN, SVM, Logistic Regression, and Naive Bayes using Accuracy, Precision, Recall, and Macro F1-score.
- Achieved **98.89% test accuracy**, **99.00% macro F1-score**, and **99.00% precision/recall** using XGBoost, outperforming other ensemble learning models on the multi-class intrusion detection task.

Atrial Fibrillation Detection from ECG Signals | Deep Learning, Time Series Analysis, Biomedical AI

- Developed a hierarchical deep learning pipeline using the **PhysioNet/CinC 2017 Challenge** dataset (8,528 ECG recordings) to classify signals into Normal, Atrial Fibrillation, Other Arrhythmias, and Noisy classes despite significant class imbalance.
- Designed a two-stage framework combining a Temporal Convolutional Network (TCN) and EfficientNet-B2, with extensive ECG preprocessing, augmentation, class balancing, and benchmarking against Transformer and classical machine learning baselines.

- Achieved **85.98% end-to-end test accuracy**, **77.20% macro F1-score**, and **0.7485 MCC**, with the Stage 1 TCN reaching **93.22% validation macro F1-score** and Stage 2 achieving **98.19% validation accuracy**.

Chronic Kidney Disease Prediction | *Machine Learning, Healthcare AI, Flask* | (Team)

- Developed a machine learning pipeline for early CKD prediction using the UCI CKD dataset, applying MICE imputation, missing-value analysis, medical range validation, clinical feature engineering (including eGFR and comorbidity scores), and feature selection.
- Benchmarked and hyperparameter-tuned XGBoost, CatBoost, Random Forest, Decision Tree, Gradient Boosting, Extra Trees, SVM, Logistic Regression, and Bernoulli Naive Bayes across baseline, feature-engineered, and feature-selected datasets.
- Designed a model risk assessment framework to detect overfitting by comparing validation and test performance, rejecting models with **100% test accuracy** due to high deployment risk and selecting XGBoost (**98.75% accuracy**) for superior generalization.
- Integrated the final model into a Flask web application with an HTML/CSS/JavaScript frontend for real-time CKD prediction and interpretable clinical decision support.

WORK EXPERIENCE

Research Intern

June 2025 – July 2025

Indian Space Academy (ISA)

- Developed the **Age of the Universe Estimation** project using the Pantheon+SH0ES Type Ia Supernova dataset to estimate the Hubble parameter and compute the age of the universe through cosmological data analysis and numerical methods.
- Worked extensively with astronomical datasets, FITS files, data cubes, and the Astropy ecosystem for scientific data processing, visualization, and analysis in Python.

AI Content Intern

May 2025 – June 2025

FlyBrand Digital Solutions

- Explored AI model development workflows using Google Teachable Machine and Roboflow by creating image, audio, and computer vision models for posture recognition, voice-command classification, and PCB defect detection.
- Produced technical training demonstrations documenting model creation, dataset preparation, training, evaluation, and deployment workflows for educational AI content.

TECHNICAL SKILLS

Languages: Python, Java, C/C++, SQL, JavaScript, TypeScript, R, HTML/CSS

AI & Machine Learning: PyTorch, TensorFlow/Keras, Scikit-learn, XGBoost, CatBoost, Deep Learning, Machine Learning, Computer Vision, NLP, LLMs, LangChain, Prompt Engineering

Data Science & Scientific Computing: Pandas, NumPy, Matplotlib, OpenCV, Astropy, NeuroKit2, music21, FITS File Processing, Jupyter Notebook

Frameworks & Web: FastAPI, Flask, Next.js, React, Tailwind CSS

Databases & Tools: PostgreSQL, MongoDB, Git, VS Code, Cursor, Claude Code, Antigravity, OpenSheetMusicDisplay (OSMD), Audiveris